

# Electrical Infrastructure Monitoring: Case of NTDCL's 500kV Network Insulator Detection With YoloV8

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**ABSTRACT** High voltage electrical infrastructure inspection requires condition monitoring of transmission line assets to avoid any possible failures or emergency. Detection of insulators in strings is linked with electrical infrastructure monitoring pertaining to the insulator fault classification. The dataset widely available for insulator monitoring are either synthetic, lab created or publicly not available. In this paper, an indigenous dataset is created using Autonomous Aerial Vehicles (AAV) technology, capturing images in diverse topographical ambience across different transmission lines/circuits managed by National transmission and dispatch company ltd. in Pakistan. For detection of insulators in string, object detector model You Only Look Once-version 8 (YOLOv8n) is trained on created dataset of 3618 images, 603 being original and other augmented, after preprocessing and augmentation techniques were applied. The model's performance is up to the mark with accuracy of 92%. The precision and recall being 0.95 and 0.90 respectively, whereas F1 score of the model peaked at 0.95 at confidence level of 0.652.

**INDEX TERMS** Condition monitoring, insulator detection, indigenous dataset, You Only Look Once (YOLO), NTDCL Pakistan.

## I. INTRODUCTION

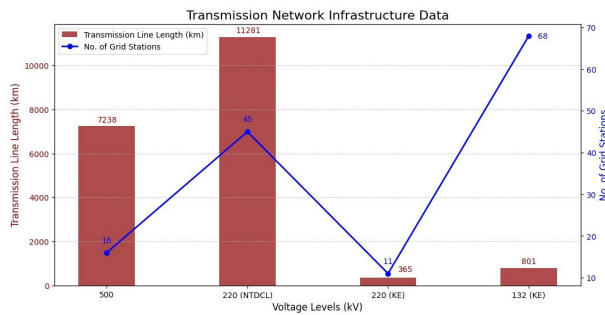
AMONG the other critical infrastructures of modern developed or developing societies, electrical power systems are crucial ones. These infrastructures contain assets services and systems in them that develop and support economic, commercial, lifesaving and social activities [1] In Electrical infrastructure, along with other components like power generation and distribution, bulk power transmission is the main networking through grid connected conductors that transfer power at different voltage levels [2], [3].

In Pakistan, national grid, i.e., 500 kV voltage level is solely operated by National Transmission and Despatch Company Limited (NTDCL), 220 kV voltage level transmission networks are operated by NTDCL and Karachi Electric (K.E.) and 132 kV voltage level transmission lines are operated by different distribution companies DISCOs, KE and

Sindh Transmission and Despatch company (Pvt.) Limited STDC [4]. The Transmission Network Infrastructure Data as published by National Electric Power Regulatory Authority NEPRA in 2020 is given in figure 1.

The paper focuses on 500 kV insulators, because the tripping on 500 kV transmission line due to insulator failure is capable of long power outages at 220 kV and 132 kV lines. So, the transmission line length of 500 kV network is 7238km with 16 grids in Pakistan containing thousands of insulators and the locations are highlighted on the map in figure 2.

Currently NTDCL Pakistan has started a pilot project for the inspection of transmission lines in Pakistan that focuses on the modern trends used for monitoring the transmission line assets, but the practice has not been finalized to be implemented in NTDCL as whole [7]. Condition monitoring of insulators is so crucial that about 70% of the faults occurring



**FIGURE 1.** Transmission network infrastructure data as per NEPRA 2020 [5].



**FIGURE 2.** 500kV grid station locations in pakistan [5].

in transmission lines are due to the faulty insulators although the cost of insulators during transmission line erection is merely 5% to 8% [8]. Conventionally, human patrolling is carried out as a part of condition monitoring and to detect the faulty insulators which is inefficient and human resource wastage. In these processes workers need to exert huge effort to go along the transmission lines or see the line through binoculars [9]. Thus, the role of autonomous aerial vehicles (AAV) is unparallel when monitoring the condition of insulators exposed to harsh environments during all seasons [10].

AAVs provide several advantages over traditional inspection methods. Looking at the far side of the insulator string is impossible through binoculars from the ground. So it becomes easy for the AAVs to give a look of the hard to see parts of the insulator strings especially the top of the very first insulator that relates to tower through pylons. AAVs can be equipped with especial thermal cameras to detect abnormally heating points due to leaking currents through the metallic parts supporting the insulator strings. This can enable the monitoring system to early identify the potential prone to fault parts of line [11].

Manual patrolling takes lot of human power and operational cost, and it can be avoided to reduce the time

and resources of the power transmission companies, hence improving the efficiency of the overall condition monitoring system. Along with improved efficiency, AAV-based monitoring systems are safe as they eliminate the need for availability of human power near the transmission towers thus these systems reduce the risk of any incidents due manual patrolling through mountainous paths, water channels etc. [7].

Due to fore written reasons, with the progress of AAV technology and image processing ability in recent years there is a gradual shift from manual patrol methods towards AAV patrol strategies. The most obvious advantage of this shift is the cost savings, but there were other benefits like greater efficiency and safely during inspections. The typical and AAV-based inspection/patrol to analyze the condition of insulators are depicted below in figure 3 below.

Thus, the AAV based monitoring systems seem to be the progressive shift in monitoring the transmission line assets, creating one of the efficient monitoring systems and last but not the least reducing potential threats and risks to lives of the workers that work in traditional monitoring systems [12] [13], [14].

The flow of this research paper is such that section II gives brief about the non-AI and AI-based work that has been done with respect to the insulator detection and condition monitoring. Section III is all about the methodology followed to achieve the results such as collecting the data through AAV to create the dataset, preprocessing the images before training the deep learning model, data augmentation and annotation tools along with the description of YOLOv8 model and its architecture. Result matrices of the YOLOv8 are discussed in section IV and section V is spared for concluding remarks with section 6 to acknowledge the helping hands for this research work.

## II. RELATED WORK PERTAINING TO INSULATOR DETECTION AND CONDITION MONITORING

The suspended insulator strings through pylons are forced to the inclination due to gravitational pull and the diversion or passing of the transmission lines through different terrains such as forests, urban areas, rivers, deserts, mountains, and green fields yields in complex backgrounds of all possible types when photographed through AAVs. These photos are subject to different challenges related to lighting effects and causing occlusion, blurriness, and interference from nearby objects. In addition to this, the suspended insulators are made up of different materials like glass, ceramic, and composites while possessing different shapes and creepage distance due to varying voltage levels of power transmission. All these complexities contribute some level of difficulties in their detection in real world problems [13].

These insulator strings have a significant role in power transmission from generation site to load center, these insulator strings not only insulate the bare conductors from grounding through towers but also provide mechanical support for suspended bundled conductors [15]. The power line

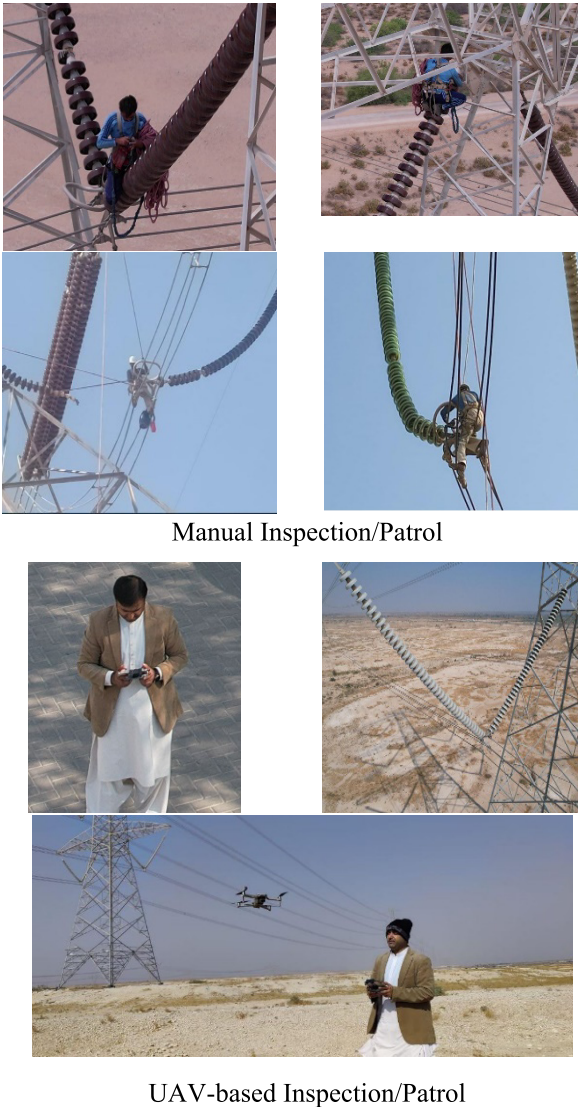


FIGURE 3. Mode of overhead line insulators inspection/patrol.

insulator failure shall cause the overall power line tripping. Conventionally to detect the insulator faults, monitoring is carried out through patrolling teams that traverse the transmission lines physically via using binoculars. This human patrolling is resource intensive and inefficient at the same time, demanding significant effort and labor [8].

The real revolution comes in this field with the advancement of intelligent inspection via visual images, especially for anomaly detection. Artificial intelligence (AI), especially deep learning, has been instrumental in this transformation, offering state-of-the-art results [16]. The evolution of deep learning theory and its application in image processing techniques have significantly enhanced object detection accuracy, surpassing conventional methods and providing substantial advantages for the effective monitoring of OHLI [17]. In 2012, deep learning algorithms gained significant traction, leading to numerous studies that employed these advanced

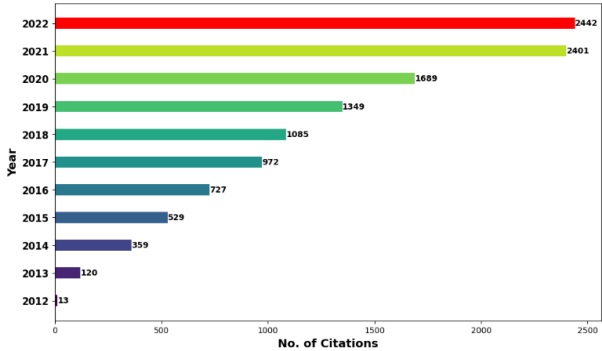


FIGURE 4. Insulator defect detection trend via the Web of Science database.

object detection techniques for identifying defects in power line insulators. As illustrated in figure 4.

There has been a notable upward trend in research focused on insulator defect detection utilizing deep learning methods. Particularly in 2022, research output surged dramatically, reaching a total of 2,442 publications, which is same number of publications as from year 2021 [13].

Researchers in [18] introduced a method for localizing insulator strings in aerial images captured by AAVs, using color filtering in the HSV color space and texture matching. The process involves converting RGB images to HSV, filtering the H and S components to segment insulators from the background, and using texture features from non-overlapping windows to refine the segmentation. This approach effectively identifies insulator outlines, reducing background interference and improving accuracy detection in complex environments. The method demonstrates significant advantages over traditional RGB-based techniques, particularly in outdoor settings with varying backgrounds and lighting conditions.

To pinpoint insulators in the aerial images of transmission lines, an elegant approach is introduced in [19]. The research is revealed with two consequent steps, first filtering the colors of aerial images in HSV space followed by the texture match. The insulator happens to be separated in aerial image very well when researchers use hue and saturation by converting the aerial image to HSV color space transformation technique. The further cluttered background is cleaned up via dividing the image in small sections and analyzing texture features. The better part is that this technique works well where lighting and background varies the most. This outperforms all techniques that rely on RGB color analysis, thus making transmission line monitoring more efficient and accurate.

AI-based insulator detection techniques leverage deep learning and computer vision to revolutionize power infrastructure monitoring. Using algorithms like CNN, SSD, YOLO etc. and drone-mounted cameras, these methods not only offer faster, safer, but more accurate inspections than traditional approaches. The AI systems can rapidly analyze

thousands of images, detecting subtle defects often missed by human inspectors, thus enabling proactive maintenance, and improving overall power distribution safety [20].

This study in [21] proposes an enhanced Faster R-CNN model for insulator detection, addressing the limitations of the VGG16 feature extraction network. The authors substituted VGG16 with ResNet50 together with a Feature Pyramid Network (FPN) to enhance the detection rate of the insulators especially if occluded or small. They also improved the standard deviation of the scales and reduced the number of anchors on the RPN and incorporated Soft-NMS to decrease the rate of missed detections. The authors reported that the improved model had the following detection precision; Detection precision = 98.67%. It can be concluded that the accuracy of the described method is 67% which is a little higher than of YOLOv3 and SSD networks.

In [22], the authors proposed an insulator detection technique using the faster R-CNN with improved modifications like the GIoU for accurate localization and repetition of NMS as Soft-NMS for the apparition of occlusions. Specifically, aerial images are used in the study, which improves the model's adaptability to the different scales of insulators and complex background. Experimental outcomes also revealed interesting performances in the detection accuracy where the method proposed in the paper yielded an AP value of 94.2%. This is more than traditional models like Faster R-CNN, SSD, and YOLOv3.

In [23], Pha et al. suggested an enhanced Faster R-CNN for detecting insulators; the authors highlighted its ability in dealing with scales, aspect ratios and occlusions. The study used improvements on the RPN and the NMS that provided great detection performance. With the new model, better results were obtained in terms of accuracy and precision where the model accurately identified occluded insulators and those with the different aspect ratios. It can be concluded that the proposed method obtained AP value equal to a 91.21%. The proposed method has better mAP values while outcompeting R-FCN and SSD models.

Insulators detection using Singe Shot MultiBox Detector (SSD) is described in paper [24] of Xiren Miao et al. To improve the detection accuracy the study uses the two-stage fine-tuning approach. First, the COCO model is trained with many aerial images to obtain a simple insulator model. Then, this model is again upgraded through specific datasets considering insulator types and situations. The method enhances the detection accuracy and speed so that the localization of insulators in complicated backgrounds attains enhanced precision irrespective of the made material of the insulators – porcelain or composite.

Another study presented in [25], where Xu and his team develop an insulator detection approach based on the SSD algorithm, the algorithm emphasized the better ability of detecting insulator in complex background and with low computational consumption. The method therefore divides the output space that includes the possibility of bounding boxes to default boxes which have different aspect ratio and scale. For

the training and testing of the study, the researchers employ a large database of visible light images, and the findings from the experiments prove that the SSD-based system is efficient in detecting small-sized insulators amid many similar objects. According to the studies, the method yields better results compared to the classical approaches as regards detection speed and accuracy is concerned.

An automatic insulator detection system is proposed by Tan in research [26] includes SSD and DenseNet for better classification. The paper also deals with the lack of data in two-stage data augmentation that allows to have a sufficient amount of training data. The proposed procedure provides an outstanding accuracy rate and a high recall rate to identify insulators in different complex backgrounds. The integration of DenseNet has the positive effect on enhancing the capacity of the system in the classification of normal and defective insulators.

In [27], Yang and Wang put forward a way of identifying insulators on transmission lines using few training samples. The method uses the YOLOv5, a deep learning-based object detection method, to detect insulators from images having natural background. The significant contribution of this study is the proposition of a small-sample learning technique that involves choosing small samples from an expansive database to train the model and, therefore, minimize the requirement of labelled data. The proposed method was tested on two public insulator image datasets, demonstrating an improvement in recognition rates by 3.30% and 2.87% compared to the original YOLOv5 when trained with only 30 samples. This approach enhances the generalization ability of the YOLOv5 model under small-sample conditions, making it more practical for real-world applications where labeled data is scarce.

Concluding the above given discussion, a comprehensive table is given below regarding the dataset and algorithm used to detect the insulators.

### III. METHODOLOGY

This section is divided into sub sections consisting of information regarding the dataset, deep learning algorithm used to train on the dataset for insulator detection and the hardware used to run the algorithm.

#### A. DATASET

The subsection gives comprehensive details regarding the creation of indigenous dataset pertaining to porcelain insulators of 500 kV transmission network of NTDC Pakistan. NTDC uses 160 kN and 120 kN fog type porcelain insulators on 500 kV transmission network. 160 kN insulator (Brown or White) is used in suspension strings and 80 kN insulators are used to handle jumper conductors. The number of insulators used in string depends on the area and ambience the transmission line traverses. A maximum of 43 insulators per string and a minimum of 28 insulators per string are used in 500 kV transmission Network.

Dataset is created via capturing pictures directly or through videos from DJI Air2s drone. In study [30] researchers while

TABLE 1. AI-based related works pertaining to insulator detection.

| Author & Year             | Techniques/Method                                                        | Dataset                                                      | Results                                      |
|---------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------|
| Zhao et al. 2019 [28]     | Faster R-CNN with Improved Anchor Generation and enhanced NMS.           | 2535 IR and 7343 labeled images                              | AP= 81.8%                                    |
| Lu et al. 2021 [22]       | Improved Faster R-CNN with generalized IoU and Soft-NMS.                 | Chinese Power Line Insulator Dataset of 600 insulator images | AP of Faster R-CNN + GIoU: 93.34%            |
| Wafi et al. 2021 [29]     | Single-stage CNN with spatial Pyramid Pooling (SPP) blocks.              | 1690 images of different dataset                             | AP= 90.31%                                   |
| Ji et al. 2023 [21]       | Improved Faster R-CNN with ResNet50+FPN and increased anchors.           | 1500 original with 3 times augmented images                  | 98.67% precision and 92.49% recall           |
| Gao et al. 2022 [23]      | Improved Faster Region-Convolutional Neural Network (R-CNN)              | 115 IR images                                                | AP= 91.21%                                   |
| Xu et al. 2018 [25]       | Single Shot MultiBox Detector (SSD)                                      | 1308 visible light images                                    | Precision= 0.95 and Recall= 0.92             |
| Xinqiang et al. 2021 [26] | Single Shot MultiBox Detector (SSD) with DenseNet classification network | 1308 actual +1308 data enhancement images                    | Precision= 93.75%, Recall= 90% and AP 94.12% |

TABLE 2. Size and diversity of indigenous dataset.

| Task       | Diverse Topographies (Actual images →Augmented Images) |              |                   |                  |
|------------|--------------------------------------------------------|--------------|-------------------|------------------|
|            | Barren Lands                                           | Green Fields | Mountain terrains | Total (603→3618) |
| Training   | 280→1400                                               | 90→450       | 113→565           | 483→2415 (80%)   |
| Testing    | 32→160                                                 | 18→90        | 10→50             | 60 →300 (10%)    |
| Validation | 27→135                                                 | 16→80        | 17→85             | 60 →300 (10%)    |

analyzing the effect of EM field of 500 kV line on AAV have concluded through experiment that when the AAV is 1m or more away from the wire, then bar-plate gap discharge will not occur. Thus, minimum 1.5m distance has been maintained. The dataset sources from following transmission circuits of NTDC.

- NKI-Jamshoro Circuit (Single Circuit)
- Dadu-Jamshoro Circuit (Single Circuit)
- CPHGC-Jamshoro Circuit (Double Circuit)
- K2 K3- Jamshoro Circuit-1 (Double Circuit)

A total of 603 original healthy insulators images have been acquired through AAV flights which have been augmented such that 5 images have been acquired for single original image, thus total images become six times the original image, i.e., 3618 images in dataset of healthy insulator images. The size and diversity of the dataset has been shown in Table 2. The topographical diversity is shown in fig 5.

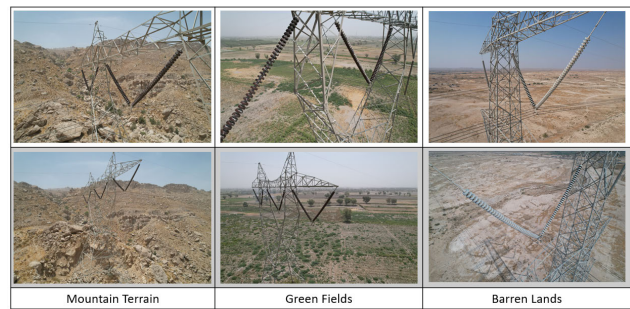


FIGURE 5. Sample images of topographical diversity.

TABLE 3. Image enhancement techniques and their advantages.

| Process                    | Technique                                                   | Advantage                                                                                                                                                                                                                   |
|----------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Color Space Transformation | Hue (Color) Saturation (Intensity) Value (Brightness) (HSV) | Converting the RGB image to grayscale or other color spaces like HSV may be beneficial before applying enhancement techniques. This can help isolate intensity information and make feature extraction more effective [31]. |
| Contrast Enhancement       | CLAHE – Contrast Limited Adaptive Histogram Equalization    | Enhances the global or local contrast, making insulators more distinguishable from the background [32].                                                                                                                     |
| Denoising                  | Bilateral filtering                                         | Reduces noise while preserving edges, which is critical for keeping the structure of insulators intact [33].                                                                                                                |
| Sharpening                 | High-pass filtering                                         | Enhances the edges of the insulators, which can help in better localization by the YOLO detector [33].                                                                                                                      |

B. PREPROCESSING THE IMAGES

For aerial images of insulator strings on transmission lines, there are various challenges such as variable lighting conditions, occlusions, background clutter, and possibly noise due to the imaging process or atmospheric conditions. Pre-processing and noise reduction can significantly enhance the quality of such images before these images are passed towards a detection algorithm like YOLO.

Table 3 below explains the preprocessing techniques employed in this research work before training the model on indigenous dataset and their advantages to enhance the features of the aerial images.

C. DATASET AUGMENTATION AND ANNOTATION

The 5 techniques used for the augmentation without changing the real being of the aerial image of insulator string are given below.

Rotation: This technique is applicable to the aerial views of insulator strings because the angles used in this method

mimic the positions of a camera during the creation of the images. Real-world insulators are observed from different perspectives at times, and though translating the images slightly (say  $\pm 30$  deg), does not affect the core characteristics of insulators, applying these small transformations can enhance the model's generalization [27].

**Brightness adjustment:** Modification of the brightness of aerial images is a form of augmenting the images that is quite realistic, as it mimics the changes in lighting that normally occurs when inspecting aerial images. It may assist the model to assign to different time of day or weather without changing the structural details of the insulator strings [27].

**Image fogging:** Since simulating fog entails copying real-life situations that may be found in the atmosphere during the time of inspection by the inspectors, it is reasonable to use the addition of fog to the aerial images as an acceptable method of augmentation. This technique can enhance the model's capability of recognizing insulators under poor visibility without altering the real content of the insulator strings [27].

**Horizontal flipping:** This technique is appropriate for aerial images of the insulator strings because the technique does not alter the strings' basic composition or orientation. Due to horizontal flipping, insulators can be seen from both sides in real-world scenarios and thus, it can improve the generalization capability without degrading it [34].

**Saturation:** In this case, making small corrections to the color balance, hue, or saturation of aerial images will help the model to produce more accurate results in different lighting conditions with different cameras. This technique is appropriate if used moderately since it does not alter the physical layout and configuration of the insulator strings but assists in mitigating for differences in image capture tools and conditions [34].

These augmentation techniques are suitable for aerial images of insulator strings because they simulate real-world variations in viewing conditions, lighting, and atmospheric effects without altering the fundamental characteristics or spatial relationships of the insulators [27], [34].

Dataset annotation is done through well-known online platform Darwin's v7labs via polygon tool with only one class. Some sample images are shown in fig 6 below.



FIGURE 6. Single class image annotation.

TABLE 4. Details of the YOLO algorithm [36].

| Model Name | Year | Anchor | Framework | Backbone         |
|------------|------|--------|-----------|------------------|
| YOLOv1     | 2015 | No     | Darknet   | Darknet24        |
| YOLOv2     | 2016 | Yes    | Darknet   | Darknet24        |
| YOLOv3     | 2018 | Yes    | Pytorch   | Darknet53        |
| YOLOv4     | 2020 | Yes    | Darknet   | CSPDarknet53     |
| YOLOv5     | 2020 | Yes    | Pytorch   | YOLOv5CSPDarknet |
| YOLOv6     | 2022 | No     | Pytorch   | EfficientRep     |
| YOLOv7     | 2023 | No     | Pytorch   | YOLOv7Backbone   |
| YOLOv8     | 2023 | No     | Pytorch   | YOLOv8CSPDarknet |

#### D. YOLOv8 ARCHITECTURE

Object detection technology has evolved significantly, with single-stage models emerging as game-changers in the field. These models, including YOLO, SSD, FCOS, RetinaNet, and EfficientDet, have revolutionized the way object detection tasks are approached. Unlike their two-stage counterparts, these models streamline the process by eliminating the need for separate region proposal networks and classifiers. The beauty of single-stage models lies in their ability to detect objects directly, resulting in remarkably faster inference times. Among these, YOLO (You Only Look Once) has captured the spotlight due to its exceptional real-time object

detection capabilities. Since its inception, YOLO has undergone numerous iterations, each building upon the strengths of its predecessor [35]. The YOLO family has expanded over the years, with versions ranging from the original YOLOv1 to YOLOv8. Each iteration has brought improvements in accuracy, speed, and versatility. This evolution of YOLO models, as outlined in table 4 below, showcases the rapid advancements in object detection technology.

The YOLOv8 architecture consists of three main components: the backbone, neck, and head, each playing a crucial role in the object detection process. The architecture of the YOLOv8 is well written below and represented in figure 7.

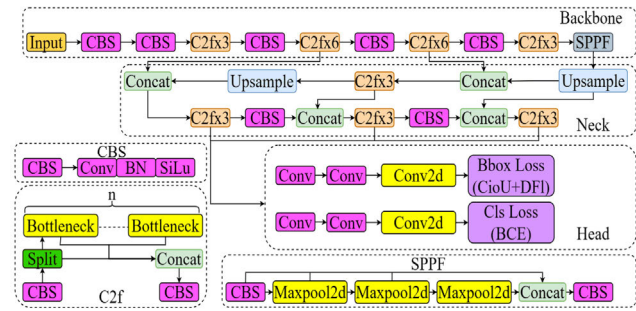


FIGURE 7. Architecture of YOLOv8 [36].

The backbone of YOLOv8 in its essence is the modification of the already mentioned CSPDarknet framework. Rather than incorporating the specifics contained in the original CSP module, it makes use of the released C2f module. To improve the flow of information inside the feature extraction network while the architecture stays simple, this module presents a gradient shunt connection. The backbone also contains a CBS module, through which information inputs are filtered by convolution, batch normalization and through the SiLU activation function. The input feature maps are normalized in an SPPF module as well where the maps are reshaped to a fixed size using pooling.

The neck of YOLOv8 is organized in PANFPN structure which combines the layouts of the top-down and bottom-up networks. This is a more subtle approach and the combining of shallow positional information with semantic features gives a richer and more filled set of features. It majorly improves the global performance of the model in object detection problems.

The head of YOLOv8 is using decoupled structure and has two branches in parallel. One branch works on the classification of objects, and the other is involved in the regression of the box around the object. Such separation makes the area monitored more precise thus increasing the chance or frequency of detection. Classification task uses binary cross-entropy loss, for the bounding box regression, the head uses distribution focal loss and complete IoU. This design choice will help enhance both the object categorization step and the localization of these objects via boxes with high precision.

#### E. PROPOSED APPROACH FOR THIS RESEARCH WORK

The approach used to carry out this research work is given in figure 8 and this is the methodology used to train the model and acquire the results on the indigenous dataset created for 500 kV transmission network of NTDC Pakistan.

### IV. RESULTS AND DISCUSSIONS

#### A. COMPUTATIONAL RESOURCES

The computational setup/machine used to conduct the training, testing and validation using indigenous dataset uses Windows 11 with Core i7 9th Gen processor, 16GB of RAM and NVidia RTX 2060 16GB graphic card.

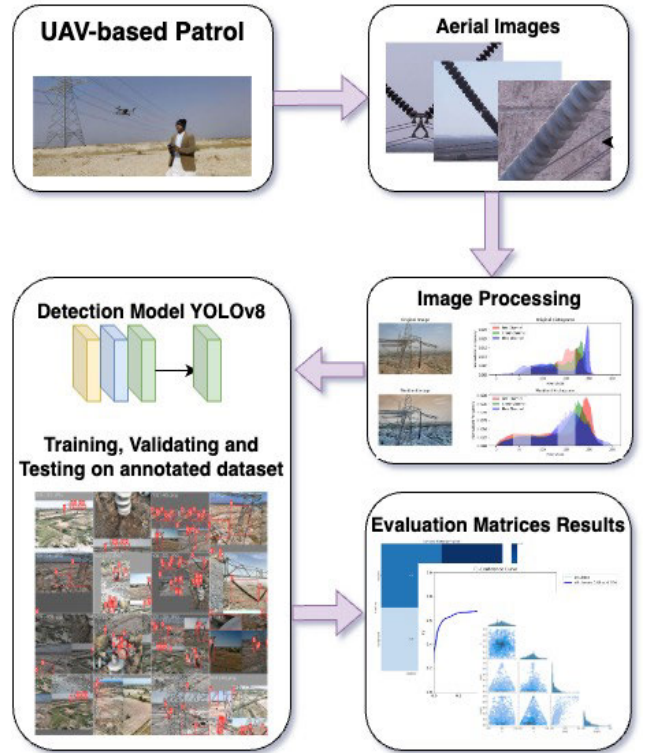


FIGURE 8. Methodology for training YOLOv8 detection model.

#### B. HYPERPARAMETERS

The hyperparameter settings used for training YOLOv8 detection model is given in table 5 below.

#### C. CONFUSION MATRIX

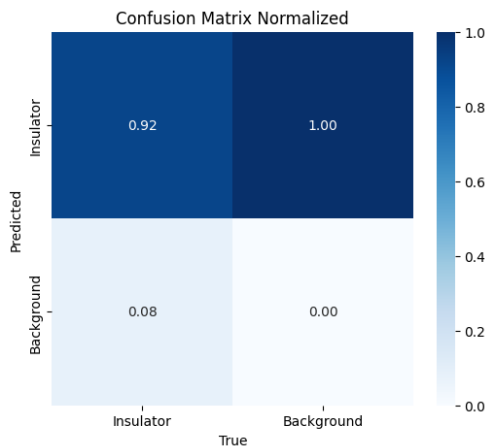
A confusion matrix is tool for evaluating the performance of AI model. The results of the YOLOv8 model received after training the model with indigenous dataset is given in fig 9. From fig 9 its evident that the model's performance on the indigenous dataset is up to the mark.

The true positive rate (TPR) for the class (insulator) is 92%, that indicates the 92 out of every 100 actual instances (insulator) are predicted correctly as insulator. The false positive rate (FPR) as presented in confusion matrix is 100%, that means 100% actual instances (background) are incorrectly predicted as insulator. The FPR is the ratio of negative instances incorrectly classified as positive (all instances that are actually background) to the actual instances (all instances when insulator is correctly predicted as insulator and background is correctly identified as background). Thus in this case YOLOv8 is employed to detect only insulator or single class "Insulator" and no prediction for the "Background" class, so all the instances that are actually "Background" will be classified as "Insulator leading to 100% FPR.

False negative rate FNR in this scenario is 8% which means that out of 100 instances, 8 instances are actual insulators are incorrectly predicted as Background. The true negative rate

**TABLE 5. Hyperparameters used in YOLOv8 detection model.**

| Hyperparameter            | Range/value |
|---------------------------|-------------|
| No. of Epochs             | 100         |
| Patience                  | 100         |
| Batch size                | 16          |
| Image size                | 640x640     |
| Workers                   | 8           |
| Optimizer                 | auto        |
| Mask ratio                | 8           |
| Learning Rate (lr0)       | 0.01        |
| Final Learning Rate (lrf) | 0.01        |
| Momentum                  | 0.937       |
| Weight decay              | 0.0005      |
| Translate                 | 0.1         |
| Scale                     | 0.5         |
| Shear                     | 0.0         |
| Auto Augment              | randaugment |



**FIGURE 9. Confusion matrix normalized.**

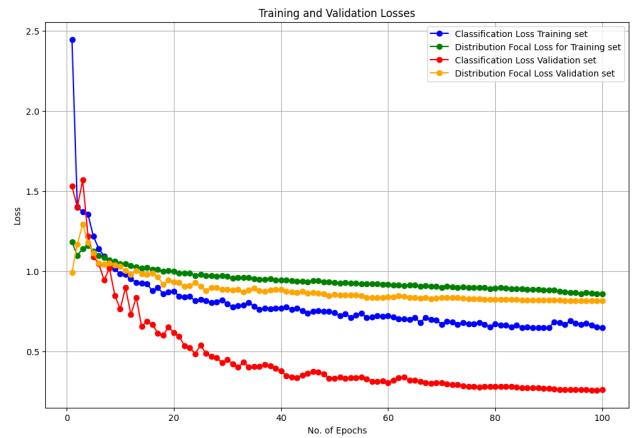
TNR is not shown in the matrix is for background class. Since the model is only detecting single class or insulator, this value is not applicable at all.

#### D. TRAINING AND VALIDATION LOSSES

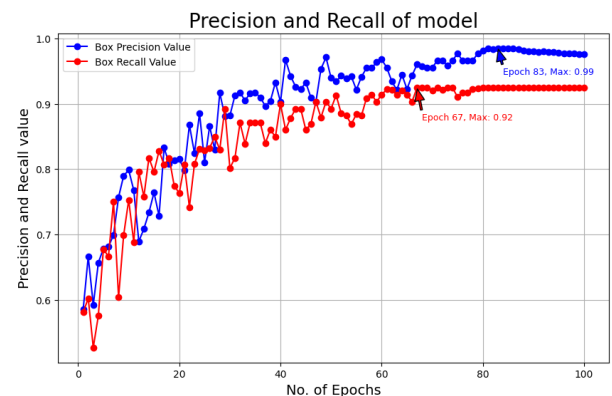
The loss for the training set and the validation set are divided into two categories, the classification loss and Distribution function loss. These losses for the detection model YOLOv8 on indigenous dataset are given in figure 10 below.

Classification loss on training dataset is given in results as train/cls\_loss, this is the measure of how well model is learning to detect the insulator during training session while classification loss on validation dataset represented as val/cls\_loss is the measure of how well the model is generalizing the unseen data. Classification loss indicates about the ability of model to correctly identify and detect the insulator. So, in both cases it is evident from fig 10 that loss is reducing with increasing epoch that indicates the better performance of model.

Distribution Focal Loss DFL is associated in choosing the hard samples and the problem associated with the class imbalance. DFL is useful especially in increasing accuracy



**FIGURE 10. Training and validation losses of YOLOv8 model.**



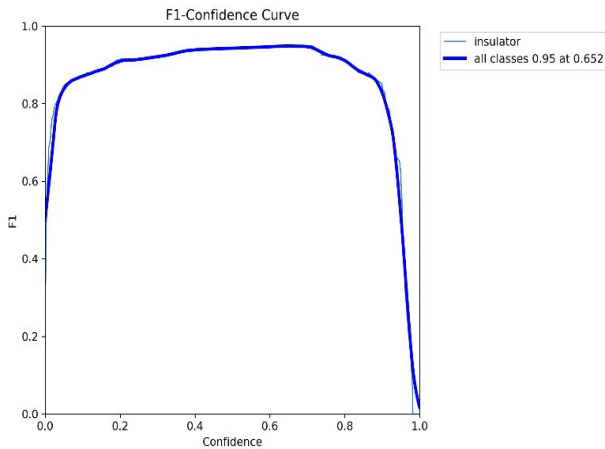
**FIGURE 11. YOLOv8's precision and recall values.**

in the localization of the object and especially when dealing with objects at different scales. From the observed trends in the DFL losses, it can be deduced that these losses reduce over epochs for both training and validation phases meaning that the model is able to learn and perform better on the insulator's detection.

#### E. PRECISION AND RECALL

Precision defines the accuracy of positive predictions. In the case of object detection it indicates how many of the successfully located items (true positives) refer to all the items detected by the model (true positives plus false positives). It can also be noted from fig 11 that as the training continues, precision tends to rise, initially starting from around the 0.585 in the first epoch and reaching 0.95 in later epochs, which suggests this improves the model's true positive rate in terms of detections.

Recall is used to measure the model's capability of identifying all relevant instances. In object detection it is the ratio of the correctly predicted objects (true positives) to the total number of present objects (true positives+ false negatives). From fig 11, it is pertinent that recall is also gradually increasing in relation to the training epochs, beginning with



**FIGURE 12.** YOLOv8's precision and recall values.

approximately 0.58065, it significantly increases to 0.90 in latter epochs which means that there is an increase in the ability of the model to determine a greater portion of true insulators in the images.

#### F. F1-CONFIDENCE CURVE

F1-Confidence curve, which is used to visualize the relationship between the F1 score and the confidence level (precision) of an insulator detection model. The curve typically starts at a lower F1 score, increases to a peak, and then decreases. This relation indicates how the model's performance varies with different confidence thresholds. The graph in fig 12 indicates that the F1 score for all classes reaches 0.95 at a confidence level of 0.652.

#### V. CONCLUSION AND FUTURE WORK

In this study, it has been illustrated that YOLOv8 is efficient in detecting insulators when trained on indigenous dataset taken through AAV from NTDC transmission network. From the confusion matrix it could be observed that the system had a high TPR of 92% thus showing that model is efficient in detecting insulators. Precision and Recall which has shown a remarkable improvement during the training, in which Precision hit its highest value 0.95 and recall attaining 0.90. The F1-Confidence curve also supported the model's stability and high performance, achieving an F1 score of 0.95 at a confidence level of 0.652. Whereas from Section-II, it is evident that Faster R-CNN achieves precision up to 98.67% and AP values of 94.2% for enhanced Faster R-CNN models, but these results are often achieved through synthetic or less diverse datasets or with higher computational requirements. SSD-based approaches have shown strong performance particularly in detecting small and occluded objects with reported mAP values from 91-94% but SSD typically requires high computational resources. These comparative results cumulatively verify that the YOLOv8 model does prove to be accurate and reliable in insulator

detection tasks. These research findings have the potential of expanding the body of knowledge on the efficacy of deep learning methods on monitoring and maintenance of power infrastructure to enhance the efficiency of the power Transmission companies. The detection algorithm can be scaled to other voltage level insulators, i.e., 220 kV or 132 kV distribution system and other transmission system components such as conductors, spacers and dampers if the dataset is created accordingly. This study is limited to insulator detection. If detection model expanded with classification of faulty insulator, it can be proved as optimized intelligent faulty insulator detection model that can address the potential challenge of non-real insulator dataset for the deployment on edge devices.

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